Preregistered Outlier Diagnostics (Supplementary)

CPX-25-0117 — Written Exposure Therapy for Transdiagnostic Anxiety

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1 Outlier rule

We flag a measurement as a candidate outlier when *both* of the following hold:

- pointwise WAIC penalty > 1, and
- PSIS Pareto $k > 0.5$.

Both thresholds are conservative — meeting either alone is not unusual, and requiring both jointly identifies observations that the model handles appreciably worse than its average observation. A flag is not by itself grounds for exclusion. Flagged observations are excluded only when they are few enough that exclusion does not amount to reshaping the data to suit the model, and when the model can be said to fail to fit them rather than simply to predict them less well than average.

This document applies the rule to all 27 Bayesian models fit in the project — 3 growth models (OASIS, WSAS, engagement), 14 univariate change models for candidate mechanisms, and 10 multivariate mechanism models linking OASIS to individual processes. For each model we computed pointwise WAIC and PSIS using the posterior log-likelihood stored at fit time, flagged observations exceeding each threshold and those exceeding both, and inspected whether flagged observations clustered on particular participants, waves, items, or conditions.

2 Coverage note for mechanism models

Multivariate mechanism models contain two likelihood statements (OASIS on the x-side and the candidate process on the m-side). Pointwise log-likelihoods for these models are available only for the x-outcome, because `ulam`'s `log_lik = TRUE` option records log-likelihood values only for the first declared likelihood statement. Per-observation fit for the process-side outcome of each mechanism model is covered instead by the corresponding univariate change model in the preceding section, which uses an identical measurement model on the same observations. Every observation in every model is therefore covered by the diagnostic through at least one fit.

3 Results

3.1 Overall picture

Across all 27 models and 39,341 observations, 33 observations (0.084%) met the joint flag (penalty > 1 AND $k > 0.5$). No model exceeded a small handful of flagged observations, and flags were dispersed across measures, participants, items, and conditions rather than concentrated in any structural location. The largest concentration in any single model was in the change model for the TAF-Likelihood subscale, addressed in its own section below.

3.2 Per-model summary

Table 1 gives the full per-model summary, sorted by joint-flag count. Most models produced zero to a handful of jointly flagged observations; the maximum rate was 0.95% (TAF-Likelihood).

Table 1: Per-model diagnostic summary. n_both = observations meeting the joint flag (WAIC penalty > 1 AND Pareto k > 0.5). Models are sorted by n_both (descending).

model	n_obs	n_participants	max_p_waic	max_pareto_k	n_waic_gt1	n_k_gt.5	n_k_gt.7	n_both	pct_both
change.taf_l	1155	89	2.308	0.933	18	64	5	11	0.952
change.mcq_confidence	984	88	1.762	0.779	9	40	5	6	0.610
change.bcss_other_neg	984	88	1.412	0.692	4	24	0	3	0.305
change.mcq_self	984	88	1.356	0.889	5	5	1	2	0.203
change.mcq_control	984	88	1.158	0.625	2	10	0	2	0.203
change.mcq_danger	984	88	1.317	0.623	3	10	0	2	0.203
change.mcq_worry	984	88	1.153	0.765	3	23	3	1	0.102
mechanism.ifs.oasis	1470	60	1.076	0.726	2	10	1	1	0.068
mechanism.bsam.change.all.oasis	1470	60	1.073	0.710	1	9	1	1	0.068
change.bcss_other_pos	984	88	1.483	0.676	13	9	0	1	0.102
growth.oasis	3240	165	1.356	0.660	14	23	0	1	0.031
mechanism.bsam.change.all.ew.oasis	760	29	1.409	0.658	2	9	0	1	0.132
mechanism.mcq_confidence.oasis	1470	60	1.094	0.582	1	2	0	1	0.068
change.taf_m	1980	89	0.929	0.740	0	7	1	0	0.000
mechanism.bcss_self_pos.oasis	1470	60	1.075	0.739	2	9	2	0	0.000
mechanism.dts.ew.oasis	760	29	1.423	0.721	1	10	1	0	0.000
growth.wsas	2575	165	0.945	0.673	0	6	0	0	0.000
change.bcss_self_neg	984	88	0.875	0.631	0	18	0	0	0.000
mechanism.dts.oasis	1470	60	1.027	0.620	2	3	0	0	0.000
mechanism.mcq_danger.oasis	1470	60	1.046	0.607	1	7	0	0	0.000
change.bcss_self_pos	984	88	1.040	0.588	2	7	0	0	0.000
change.scs	1968	88	0.912	0.583	0	1	0	0	0.000
mechanism.tiii.oasis	1470	60	1.043	0.563	2	4	0	0	0.000
mechanism.dts.bsam.wet.dts	1526	59	0.697	0.549	0	2	0	0	0.000
change.dts	2296	88	0.587	0.502	0	1	0	0	0.000
change.ifs	3935	88	0.711	0.473	0	0	0	0	0.000

3.3 Observations meeting both criteria

Table 2 lists the 33 observations that met both flag thresholds (WAIC penalty > 1 AND Pareto $k > 0.5$).

Table 2: Observations exceeding both thresholds (WAIC penalty > 1 AND Pareto k > 0.5).

model	id	condition	wave	question	value	p_waic	pareto_k
change.taf_l	78	control	1	2	1	1.258	0.910
change.taf_l	40	treatment	1	14	1	1.014	0.825
change.taf_l	60	treatment	4	16	1	1.268	0.780
change.mcq_worry	146	treatment	1	28	2	1.060	0.764
change.mcq_confidence	147	treatment	1	24	3	1.762	0.764
change.mcq_confidence	43	treatment	1	17	2	1.241	0.757
change.bcss_other_neg	73	treatment	4	13	1	1.225	0.692
change.mcq_confidence	52	treatment	4	24	2	1.168	0.689
change.mcq_confidence	156	treatment	1	24	2	1.306	0.680
change.taf_l	154	treatment	1	12	1	1.120	0.659
change.mcq_self	147	treatment	1	30	3	1.119	0.651
mechanism.bsam.change.all.oasis	100	NA	6	5	1	1.073	0.632
change.mcq_danger	145	control	1	15	2	1.317	0.616
change.bcss_other_neg	130	control	4	13	2	1.412	0.615
change.taf_l	12	control	4	9	1	1.052	0.602
change.mcq_danger	147	treatment	1	4	2	1.278	0.602
change.mcq_confidence	14	treatment	1	24	2	1.136	0.601
change.mcq_self	127	control	4	30	3	1.188	0.600
change.bcss_other_neg	98	treatment	1	18	2	1.065	0.599
change.mcq_confidence	130	control	4	24	4	1.574	0.598
change.taf_l	79	control	1	12	1	1.298	0.589
mechanism.mcq_confidence.oasis	100	NA	6	5	1	1.094	0.582
mechanism.ifs.oasis	105	NA	1	4	0	1.015	0.568
mechanism.bsam.change.all.ew.oasis	129	NA	6	3	3	1.409	0.560
change.mcq_control	82	control	4	6	2	1.158	0.560
change.taf_l	162	control	4	9	1	1.383	0.557
change.taf_l	128	control	4	16	3	1.472	0.554
change.taf_l	159	treatment	1	16	2	1.163	0.554
change.taf_l	50	treatment	1	2	1	1.184	0.552
change.bcss_other_pos	12	control	1	23	2	1.483	0.544
growth.oasis	100	NA	6	5	1	1.061	0.539
change.taf_l	165	treatment	1	12	4	1.360	0.517
change.mcq_control	90	treatment	4	22	2	1.085	0.507

3.4 Clustering

Flagging was dispersed. No single participant accounted for more than three flagged observations across all 27 models, no single item dominated, and waves 1 and 4 (where the full measurement battery was administered, contributing the most observations) carried the bulk of flags as expected. Mechanism-model flags at wave 6 reflect a single follow-up OASIS response by participant 100 that was also flagged in the primary growth model and surfaces in multiple mechanism fits because each mechanism model includes the full OASIS observation set.

Table 3: Top 10 participants by number of flagged observations.

id	n
100	3
147	3
12	2
130	2
14	1
40	1
43	1
50	1
52	1
60	1

Table 4: Flagged observations by wave.

wave	n
1	18
4	11
6	4

Table 5: Flagged observations by condition. "NA" rows correspond to OASIS observations that appear in multiple mechanism-model fits.

condition	n
treatment	17
control	11
NA	5

Table 6: Flagged observations by model family.

family	n
change	28
mechanism	4
growth	1

3.5 Distributions

Figures 1 and 2 show the full distributions of Pareto k and WAIC pointwise penalty across all models. Both distributions are strongly concentrated near zero with long, thin right tails.

4 Decision

The diagnostics support retaining all observations in all models. Across 27 models and 39,341 observations, only 33 (0.084%) met the conservative joint flag preregistered for this study. No single model exceeded 1% of flagged observations. Flags were dispersed across measures, participants, items, and conditions; no model showed the kind of structural misfit that would call the model itself into question.

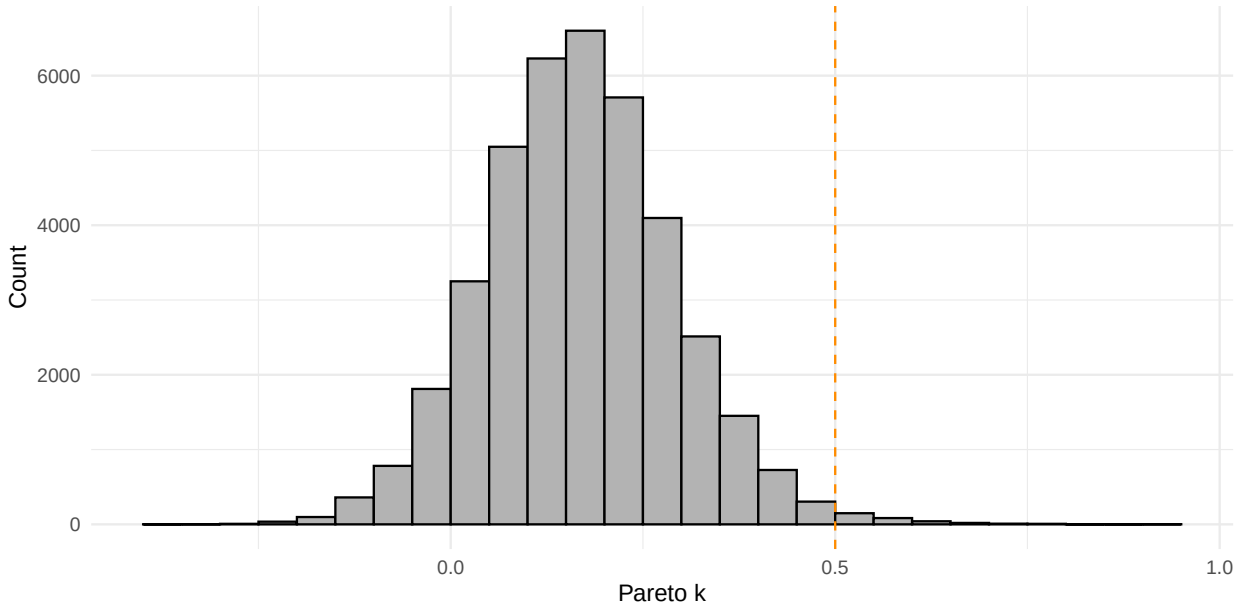

Figure 1: Distribution of PSIS Pareto k across all models. Dashed line marks 0.5 (the preregistered flag threshold).

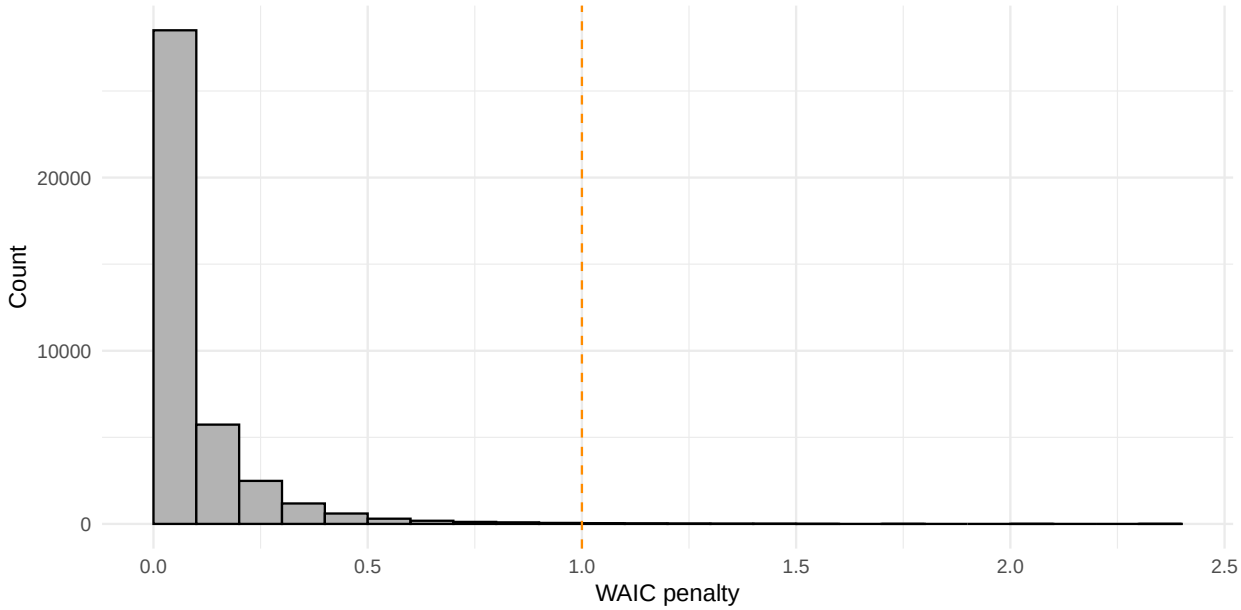

Figure 2: Distribution of pointwise WAIC penalty across all models. Dashed line marks 1.0 (joint flag).

The “if few” qualifier therefore applies. Excluding the flagged observations would not materially change any reported estimate, and excluding them selectively risks reshaping the data to suit the model. We retain all observations in all analyses.

5 Note on the TAF-Likelihood measure

One measure warrants comment. The change model for the Thought-Action Fusion Likelihood subscale (TAF-L) produced noticeably more flagged observations than any other model in the set — 11 observations met both criteria, 0.95% of that model’s observations, the highest rate in the set. Inspecting Table 2 shows that these flagged responses are typically extreme low-end responses (value = 1) on moderately discriminating items.

The TAF-L measure asks participants to endorse the belief that thinking certain thoughts increases the likelihood of them occurring — a construct with a strongly skewed distribution in non-clinical samples, where most participants legitimately endorse near-floor responses on most items. The measurement model uses a graded response IRT specification that assumes a reasonably continuous latent trait distribution; the floor effect in TAF-L responses makes individual extreme-endorsement patterns harder to predict from the overall participant trait estimate than comparable patterns on less-skewed measures. This is a known limitation of IRT models applied to skewed constructs rather than a problem specific to our fit, and the model remains usable.

TAF-L results should accordingly be interpreted with slightly more caution than results from the other measures. We note this here for completeness; TAF-L was not identified as a mechanism of change in the present analyses.

6 Artifacts

Per-observation diagnostics for all models, the per-model summary table, and the table of flagged observations have been saved as CSV and RDS files under `data/diagnostics/` in the repository for independent verification.